

Technology #52

Apply a micro■textured or hydrophilic anti■adhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial

Engineering Technology-Transfer Dossier
CONFIDENTIAL
colonization.

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply a micro■textured or hydrophilic anti■adhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

WHY DOES IT MATTER?

Post-operative endophthalmitis following intraocular lens implantation

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 11 design inputs, 14 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 11-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

canonical invention specification (with epistemic classes); engineering specification (this document, all blocks classed); design input/output graph with explicit IDs; engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification); killer-experiment definition and its EIG basis; evidence custody references and hashes; unresolved-uncertainty register; domain equation set (symbolic; sources and applicability judgments identified)

WHAT DOES THE BUYER HAVE TO BUILD?

actual geometry and detailed design; sourced engineering parameters and acceptance thresholds; physical prototypes; all verification and validation results; manufacturing process qualification; regulatory pathway determination and submissions

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 11-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Perform a controlled in vitro assay where coated and uncoated lenses are subjected to simulated eye rubbing forces; measure adhesion forces.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Mechanical athalamia due to eye rubbing triggers forward IOL displacement, causing adhesion to the anterior capsule and trapping fluid.

POTENTIAL VALUE DRIVER: Addresses: Post-operative endophthalmitis following intraocular lens implantation

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Apply a micro■textured or hydrophilic anti■adhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply a micro■textured or hydrophilic anti■adhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

```
{'equation_id': 'FLUID-001', 'name': 'Hagen-Poiseuille (laminar pipe flow)', 'expression': 'Q = (pi * r^4 * dP) / (8 * mu * L)', 'source': {'text': 'standard Newtonian laminar pipe-flow relation (any fluid-mechanics text)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'fluidics_hydraulic'}, 'assumptions': ['CSF approximated as Newtonian (valid at shunt flow rates)', 'no wall compliance effects'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions': ['CSF approximated as Newtonian (valid at shunt flow rates)', 'no wall compliance effects'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow', 'model_assumptions': ['CSF approximated as Newtonian (valid at shunt flow rates)', 'no wall compliance effects'], 'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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{'equation_id': 'FLUID-002', 'name': 'Reynolds number', 'expression': 'Re = (rho * v * D) / mu', 'source': {'text': 'standard dimensionless similarity number (any fluid-mechanics text)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'circular section; single-phase flow', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'fluidics_hydraulic'}, 'assumptions': ['transition criterion Re~2300 is a MODEL_DERIVED engineering rule of thumb'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions': ['transition criterion Re~2300 is a MODEL_DERIVED engineering rule of thumb'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'circular section; single-phase flow', 'model_assumptions': ['transition criterion Re~2300 is a MODEL_DERIVED engineering rule of thumb']}
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'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables':
[], 'mechanism_mention_count': 0}}
{'equation_id': 'FLUID-003', 'name': 'Orifice flow (sharp-edged restriction)', 'expression': 'Q = C_d * A *
sqrt(2 * dP / rho)', 'source': {'text': 'standard orifice-equation form (any fluid-mechanics text)',
'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition':
'turbulent orifice flow; C_d treated as MODEL_DERIVED until measured', 'epistemic_class': 'MODEL_DERIVED',
'judged_for_domain': 'fluidics_hydraulic'}, 'assumptions': ['C_d ~0.6-0.65 is an engineering prior, not a
measurement'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL',
'reason': 'domain template justifies this relation for the domain, but the invention text does not yet
engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables':
[], 'assumption_check': {'assumptions': ['C_d ~0.6-0.65 is an engineering prior, not a measurement'],
'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'},
'mechanism_mentions': []}, 'model_applicability': 'turbulent orifice flow; C_d treated as MODEL_DERIVED
until measured', 'model_assumptions': ['C_d ~0.6-0.65 is an engineering prior, not a measurement'],
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[], 'mechanism_mention_count': 0}}

```

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'INFECTION', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'adversarial dimension: unsupported_mechanism', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: weak_transfer', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: obvious_combination', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: prior_art', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: contradiction', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: boundary_failure', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: engineering_infeasibility', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: regulatory_incompatibility', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Must prevent microbial colonization on lens surfaces without systemic antibiotics	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Mechanical athalamia due to eye rubbing triggers forward IOL displacement	MODEM_EO
DI-003	Expected effect target	Reduced adhesion between the IOL and capsule, lower fluid retention, and more likely a decreased surface area for m	MODEM_EO
DI-004	Domain design input pattern: required flow applicability	NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: medium viscosity applicability	NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anatomical applicability	NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability	NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-008	Custodied observation: Intraocular infections	https://ui.adsabs.harvard.org/article/doi/10.1006/s001411199500141b ; hash 06a41c015e50489e34c293ae2c93af9a91cf012106	SOURCE_FACT
DI-009	Custodied observation: Primary source	https://europepmc.org/article/doi/10.1006/s001411199500141b ; content_hash 2d4d6a41c015e50489e34c293ae2c93af9a91cf012106	SOURCE_FACT
DI-010	Custodied observation: Accidental intraocular surgery	https://europepmc.org/article/doi/10.1006/s001411199500141b ; content_hash 2d4d6a41c015e50489e34c293ae2c93af9a91cf012106	SOURCE_FACT
DI-011	Custodied observation: Postoperative Capsular Bag Distension	https://europepmc.org/article/doi/10.1006/s001411199500141b ; content_hash 2d4d6a41c015e50489e34c293ae2c93af9a91cf012106	SOURCE_FACT

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Apply a micro	CONCEPTUAL	Adhesive coating definition
DO-002	pressure regulation element realizing Apply a micro	CONCEPTUAL	Adhesive coating definition
DO-003	sensing / feedback element (if closed-loop) realizing Apply a micro	CONCEPTUAL	Adhesive coating definition
DO-004	termination interfaces (patient / reservoir) realizing Apply a micro	CONCEPTUAL	Adhesive coating definition
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal	CONCEPTUAL	Adhesive coating definition
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop)	CONCEPTUAL	Adhesive coating definition
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces	CONCEPTUAL	Adhesive coating definition
DO-008	Device boundaries: patient/fluid inlet and outlet termination	CONCEPTUAL	Adhesive coating definition
DO-009	Flow/force/signal path geometry realizing the mechanism	CONCEPTUAL	Adhesive coating definition
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	Adhesive coating definition
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	Adhesive coating definition
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	Adhesive coating definition
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	Adhesive coating definition
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	Adhesive coating definition
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	Adhesive coating definition
DO-016	No closed-loop control is proposed: the invention operates as a passive open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	Adhesive coating definition
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	Adhesive coating definition
DO-018	Fixture enabling: bench flow-loop characterization across operating range — rig concept	CONCEPTUAL	Adhesive coating definition
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying transitions) — rig concept	CONCEPTUAL	Adhesive coating definition
DO-020	Fixture enabling: accelerated occlusion / fouling challenge — rig concept	CONCEPTUAL	Adhesive coating definition
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope — rig concept	CONCEPTUAL	Adhesive coating definition

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
INFECTION		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: unsupported_mechanism		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: weak_transfer		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: obvious_combination		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: prior_art		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: contradiction		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: boundary_failure		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: engineering_infeasibility		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension: regulatory_incompatibility		design control NOT ESTABLISHED	V5-H02 (requires design work)	UNKNOWN — untested candidate failure mode (Art. 2)
proximal occlusion by tissue ingrowth		inlet geometry and surface finish; anti-ingrowth surface	V5-H05	UNKNOWN — untested candidate failure mode (Art. 2)
lumen collapse under external pressure (kink)		wall thickness and reinforcement; minimum bend radius	V6-H06	UNKNOWN — untested candidate failure mode (Art. 2)
encrustation / mineral deposition		surface energy/coating selection; eliminate stagnation zones	V5-H07	UNKNOWN — untested candidate failure mode (Art. 2)
regime transition to turbulence altering calibration		operating-envelope definition; regime-margin verification	V6-H08	UNKNOWN — untested candidate failure mode (Art. 2)
check-valve seat wear opening reverse leakage		seat material and geometry; cycle-life margin; filtration	V7-H09	UNKNOWN — untested candidate failure mode (Art. 2)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
INFECTION		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: unsupported_mechanism		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: weak_transfer		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: obvious_combination		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: prior_art		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: contradiction		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: boundary_failure		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: engineering_infeasibility		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
adversarial dimension: regulatory_incompatibility		design control NOT ESTABLISHED (requires design work)	UNKNOWN - untested candidate failure mode (Art. XXV)
proximal occlusion by tissue ingrowth		inlet geometry and surface finish; anti-ingrowth surface treatment	UNKNOWN - untested candidate failure mode (Art. XXV)
lumen collapse under external pressure (kink)		wall thickness and reinforcement; minimum bend radius	UNKNOWN - untested candidate failure mode (Art. XXV)
encrustation / mineral deposition		surface energy/coating selection; elimination of stagnation zones	UNKNOWN - untested candidate failure mode (Art. XXV)
regime transition to turbulence altering calibration		operating-envelope definition; regime-matching	UNKNOWN - untested candidate failure mode (Art. XXV)
check-valve seat wear opening reverse leakage		seat material and geometry; cycle-life	UNKNOWN - untested candidate failure mode (Art. XXV)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Perform a controlled in vitro assay where bacteria are subjected to standard and design conditions	Perforated and controlled in vitro assay	perforated and standard and design conditions	NOT_PERFORMED
VF-002	run killer experiment: falsification_test	run killer experiment (EIS/Cas10.4457363932040945) data (EIS/Cas10.4457363932040945)	run killer experiment (EIS/Cas10.4457363932040945) data (EIS/Cas10.4457363932040945)	NOT_PERFORMED
VF-003	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	NOT_PERFORMED
VF-004	pressure-drop mapping vs flow (identify regime transitions)	pressure-drop mapping vs flow (identify regime transitions)	pressure-drop mapping vs flow (identify regime transitions)	NOT_PERFORMED
VF-005	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	NOT_PERFORMED
VF-006	pressure-flow mapping across full operating envelope	pressure-flow mapping across full operating envelope	pressure-flow mapping across full operating envelope	NOT_PERFORMED
VF-007	long-duration patency bench soak with particulate challenge	long-duration patency bench soak with particulate challenge	long-duration patency bench soak with particulate challenge	NOT_PERFORMED
VF-008	pressure-flow mapping across full operating envelope	pressure-flow mapping across full operating envelope	pressure-flow mapping across full operating envelope	NOT_PERFORMED
VF-009	accelerated occlusion / fouling challenge testing	accelerated occlusion / fouling challenge testing	accelerated occlusion / fouling challenge testing	NOT_PERFORMED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	Medical-grade silicone elastomer	Not Established	Not Established	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	Not Established	Not Established	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + canonical invention specification (with epistemic classes)
- + engineering specification (this document, all blocks classed)
- + design input/output graph with explicit IDs
- + engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification)
- + killer-experiment definition and its EIG basis
- + evidence custody references and hashes
- + unresolved-uncertainty register
- + domain equation set (symbolic; sources and applicability judgments identified)

YOU MUST DEVELOP:

- > actual geometry and detailed design
- > sourced engineering parameters and acceptance thresholds
- > physical prototypes
- > all verification and validation results
- > manufacturing process qualification
- > regulatory pathway determination and submissions

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'ALL sourced engineering parameter values', 'reason': 'no measurement or physical observation exists'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification	physical verification	NOT ESTABLISHED	first physical article	NOT ESTABLISHED
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Perform a controlled in vitro test	NOT ESTABLISHED	VF-001 article	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED	VF-002 article	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	bench flow-loop characterization	NOT ESTABLISHED	VF-003 article	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	pressure-drop mapping	NOT ESTABLISHED	VF-004 article	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	bench flow-loop characterization	NOT ESTABLISHED	VF-005 article	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	pressure-flow mapping	NOT ESTABLISHED	VF-006 article	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	long-duration patency test	NOT ESTABLISHED	VF-007 article	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	pressure-flow mapping	NOT ESTABLISHED	VF-008 article	NOT ESTABLISHED (to be quoted)

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-11	article for VF-009	per method	accelerated occlusion	NOT ESTABLISHED	NOT ESTABLISHED	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 11-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.